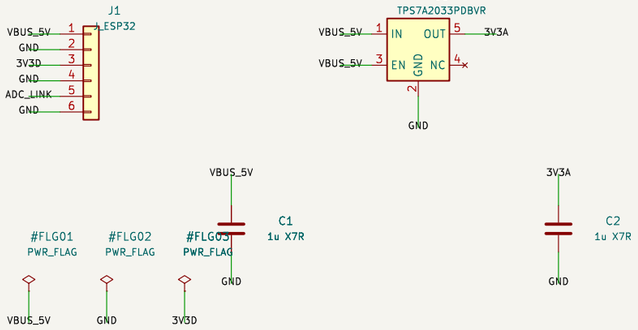
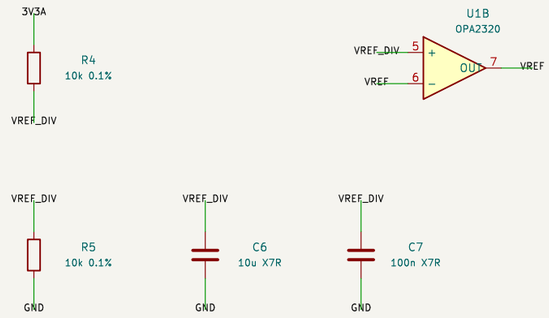
PHOTODIODE TIA / ESP32 | REV A · ENGINEERING PROTOTYPE

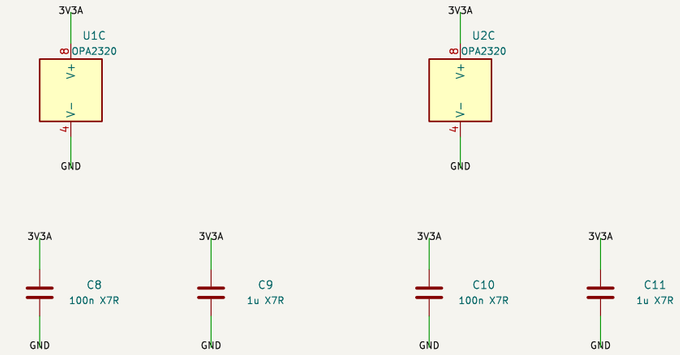
01 USB supply and quiet analog rail



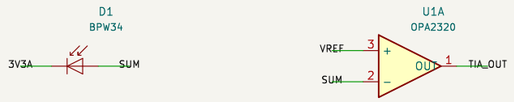
02 1.65 V reference · no output capacitor



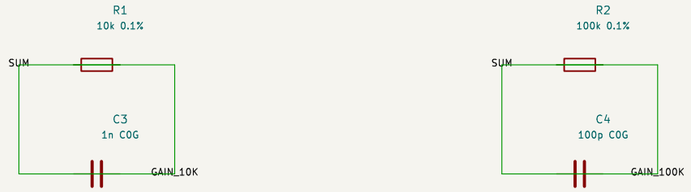
03 Local amplifier bypass · place at pins 8 / 4

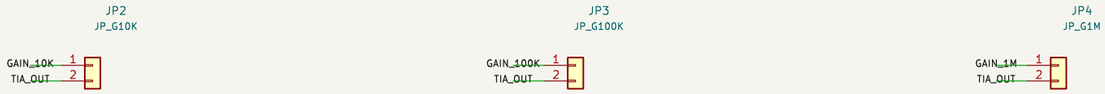


04 Reverse-biased detector and transimpedance amplifier



05 Paired gain selection · links are at the OUTPUT ends

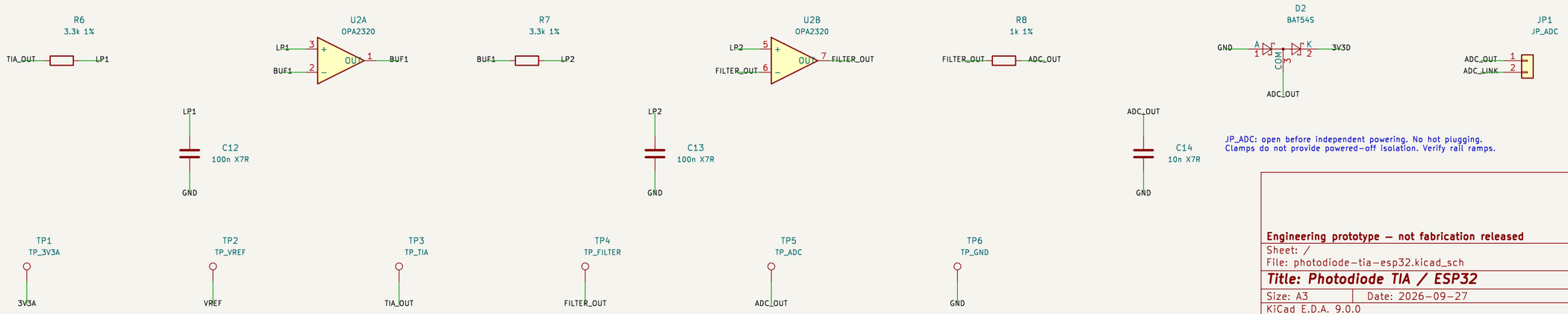




VOUT = VREF − IPHOTO × RF
Dark ≈ 1.65 V; useful full scale 0.25 V
Do not probe SUM with a conventional scope probe.

POWER OFF to change gain. Fit exactly ONE shunt. Never operate without feedback.

06 Two buffered low-pass sections · 482 Hz each · science band DC-100 Hz



JP_ADC: open before independent powering. No hot plugging.
Clamps do not provide powered-off isolation. Verify rail ramps.

Engineering prototype — not fabrication released

Sheet: /
File: photodiode-tia-esp32.kicad_sch

Title: Photodiode TIA / ESP32

Size: A3	Date: 2026-09-27	**Rev: A**
KiCad E.D.A. 9.0.0		Id: 1/1